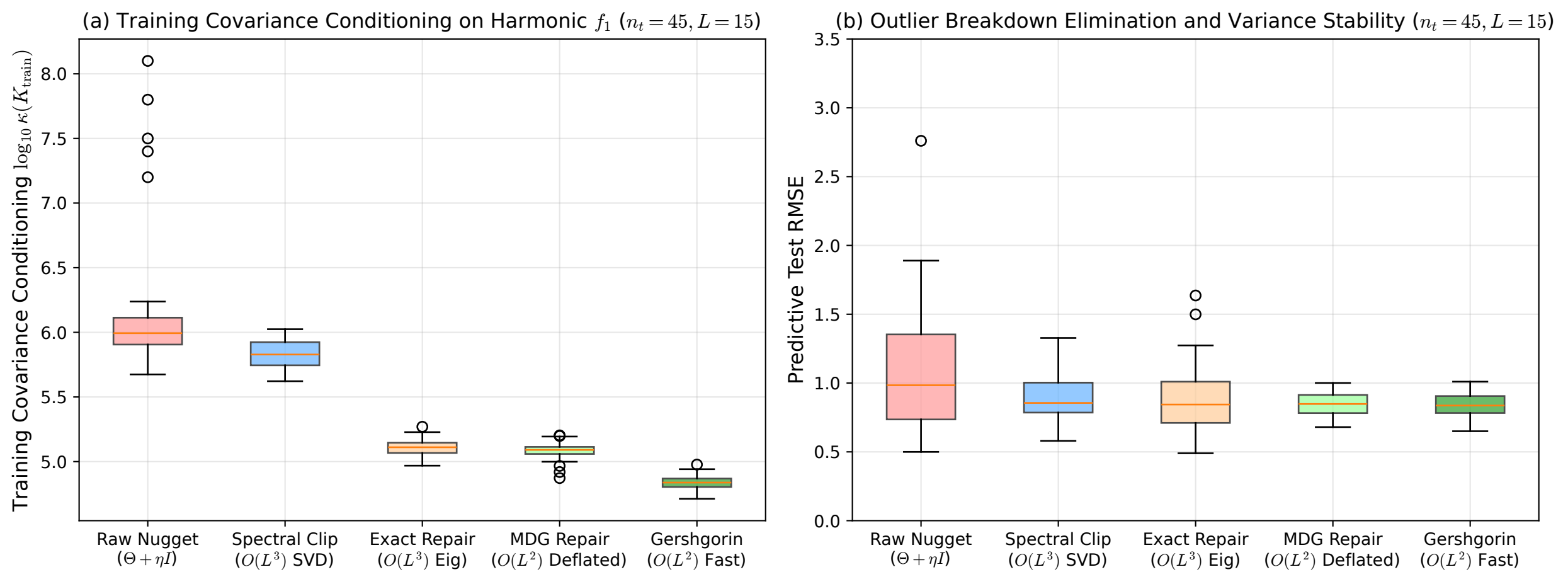
(a) Training Covariance Conditioning on Harmonic f_1 ($n_t = 45, L = 15$)
Training Covariance Conditioning $\log_{10} \kappa(K_{\text{train}})$
8.0
7.5
7.0
6.5
6.0
5.5
5.0
Raw Nugget ($\Theta + \eta I$)
Spectral Clip ($O(L^3)$ SVD)
Exact Repair ($O(L^3)$ Eig)
MDG Repair ($O(L^2)$ Deflated)
Gershgorin ($O(L^2)$ Fast)
(b) Outlier Breakdown Elimination and Variance Stability ($n_t = 45, L = 15$)
Predictive Test RMSE
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Raw Nugget ($\Theta + \eta I$)
Spectral Clip ($O(L^3)$ SVD)
Exact Repair ($O(L^3)$ Eig)
MDG Repair ($O(L^2)$ Deflated)
Gershgorin ($O(L^2)$ Fast)